greatlearning
Power Ahead

DevOps Tutorial



©Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited

greatlearning
Power Ahead

Module 1:
DevOps Overview



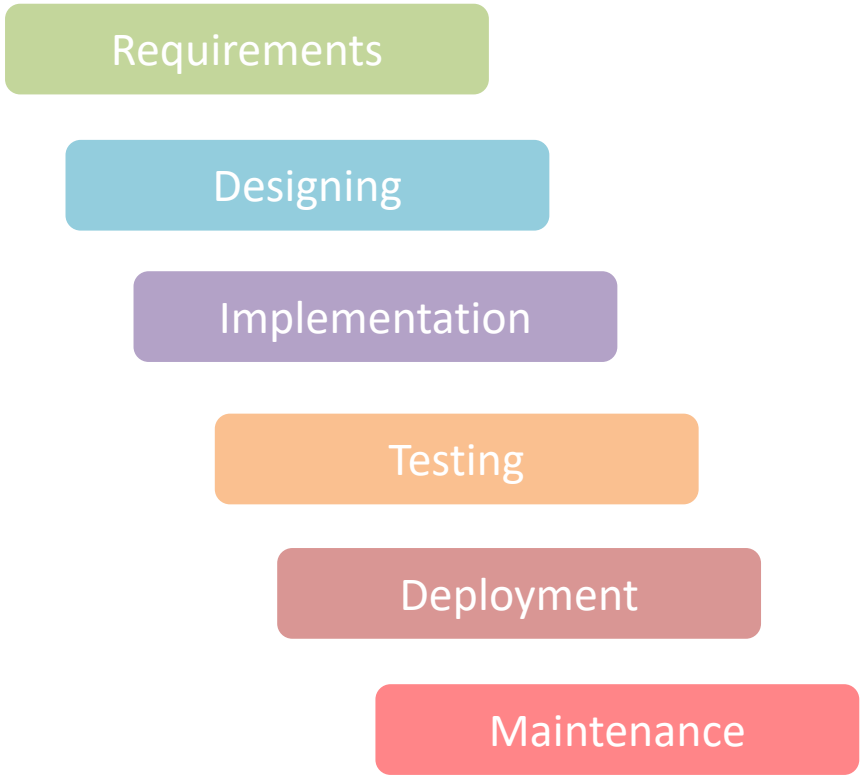
©Great Learning. All Rights Reserved. Unauthorized use or distribution pro



Why DevOps?

©Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited

Waterfall Model



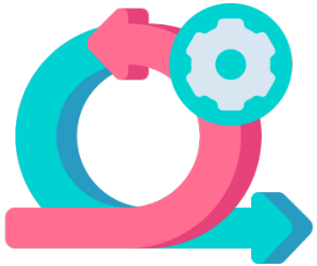
In the early year of software development when it was just coming up, waterfall model was used.

Waterfall model is a very standard model used in many different fields, not just software development.

It was very useful when the requirements were concrete and the development cycles were long.

©Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited

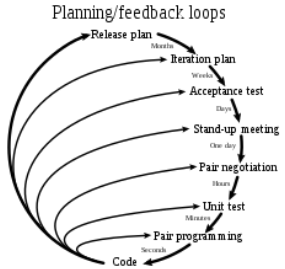
New Methodologies



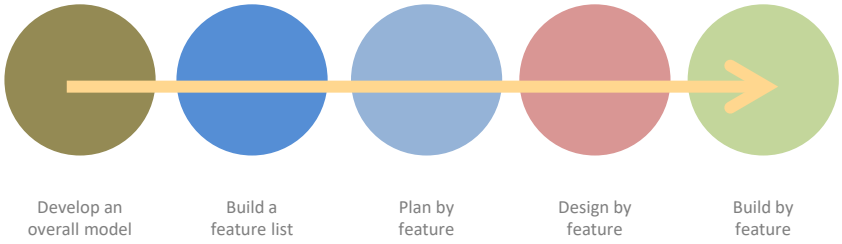
Agile



Spiral Model



Extreme Programming



Feature Driven Development



Lean Methodology

©Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited



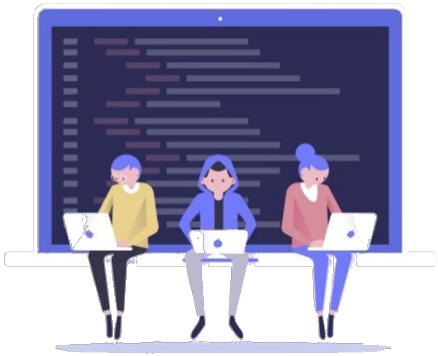
What is DevOps?

©Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited

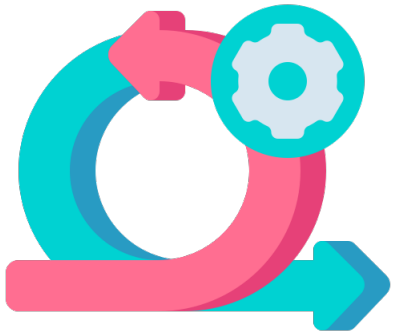
What is DevOps?



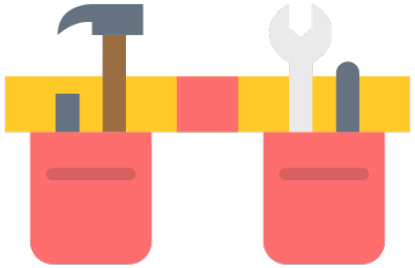
Way of software development



Values & Principle



Methodologies

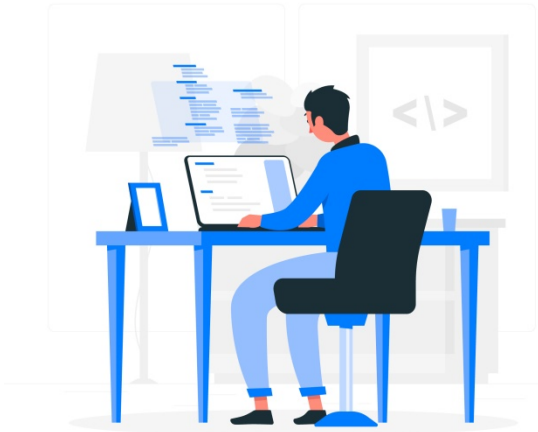


Tools

©Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited

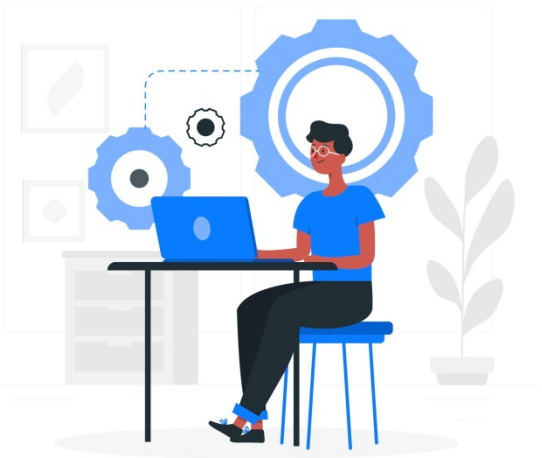
What is DevOps?

The name DevOps comes from Developer and Operations. DevOps bridges the communication gap between the software developers and the It operation teams.



Developers

DevOps



Operations

©Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited

What is DevOps?

"DevOps is a set of practices that combines software development (*Dev*) and IT operations (*Ops*). It aims to shorten the systems development life cycle and provide continuous delivery with high software quality. DevOps is complementary with Agile software development; several DevOps aspects came from Agile methodology."

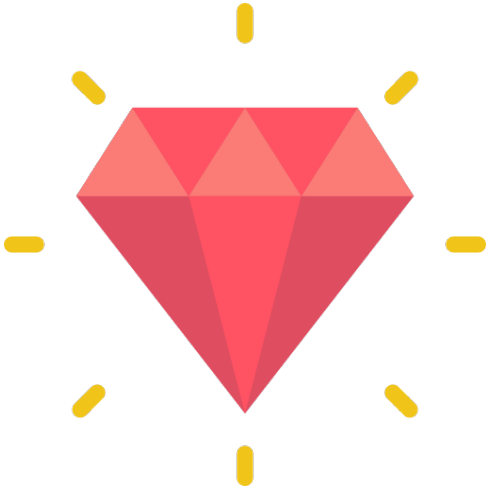


©Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited

Benefits of DevOps



Productivity

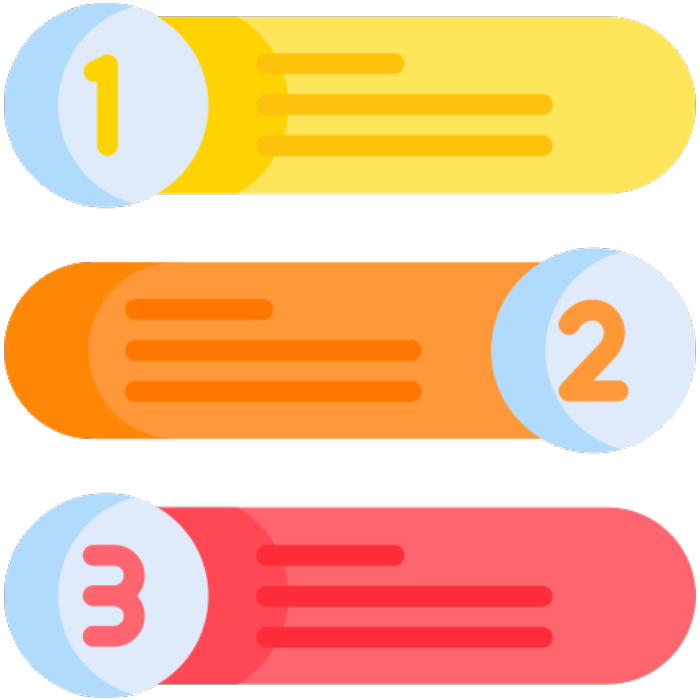


Excellence



Returns

©Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited



DevOps Phases

©Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited

greatlearning
Power Ahead

DevOps Phases

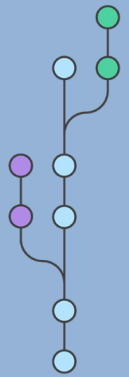
Continuous Development



Continuous Testing



Continuous Integration



Continuous Deployment



Continuous Monitoring



©Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited



Who is a DevOps Engineer?

©Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited

DevOps Engineer

- Basically a DevOps engineer is responsible for designing, maintaining the software development pipeline. He also makes sure that a software is deployed properly without any issues. A DevOps engineer knows how to automate processes and writing automation scripts. He also knows how to keep the whole infrastructure secure and robust.

- A DevOps engineer has excellent communication skills that help him to convey ideas and exchange information between different teams.

- On daily level, a DevOps engineer is responsible for makes sure scaling needs of cloud are fulfilled, optimisation, managing permissions, documentations.

©Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited

DevOps Engineer

A DevOps engineer has a wealth of knowledge such as:

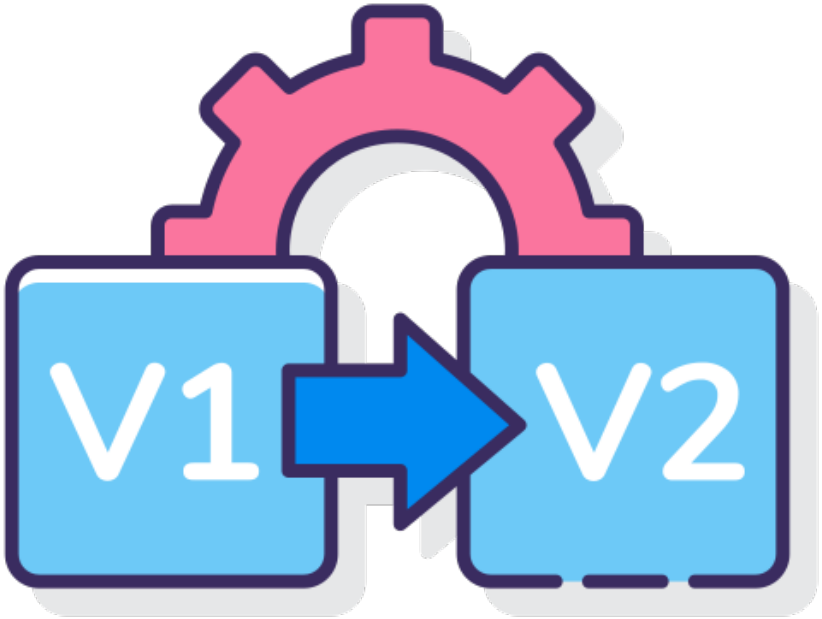
- Knowledge of Linux, Windows Operating Systems
- Database Tech
- Server Tech
- Orchestration
- Cloud tech
- Source Control
- Continuous integration & Deployment
- Automation
- Scripting
- Networking
- Excellent Communication

©Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited

greatlearning
Power Ahead



Version Control - GIT



What is Version Control?

What is Version Control?

These days when software is developed, It is not developed with the mind-set that there will only be one piece of code that will be deployed and that's it. These days smaller snippets of code are deployed in regular successions with regular feedbacks. This leads to many different versions of the code.

And that creates a need to organise the code and all of its different version of it. This is where Version Control comes in. It is a practice of managing and storing different version of a source code.

This is especially the case with Larger companies that have multiple projects and multiple teams working within it.

A1, A2, A3

B1, B2, B3

C1,C2,C3

©Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited

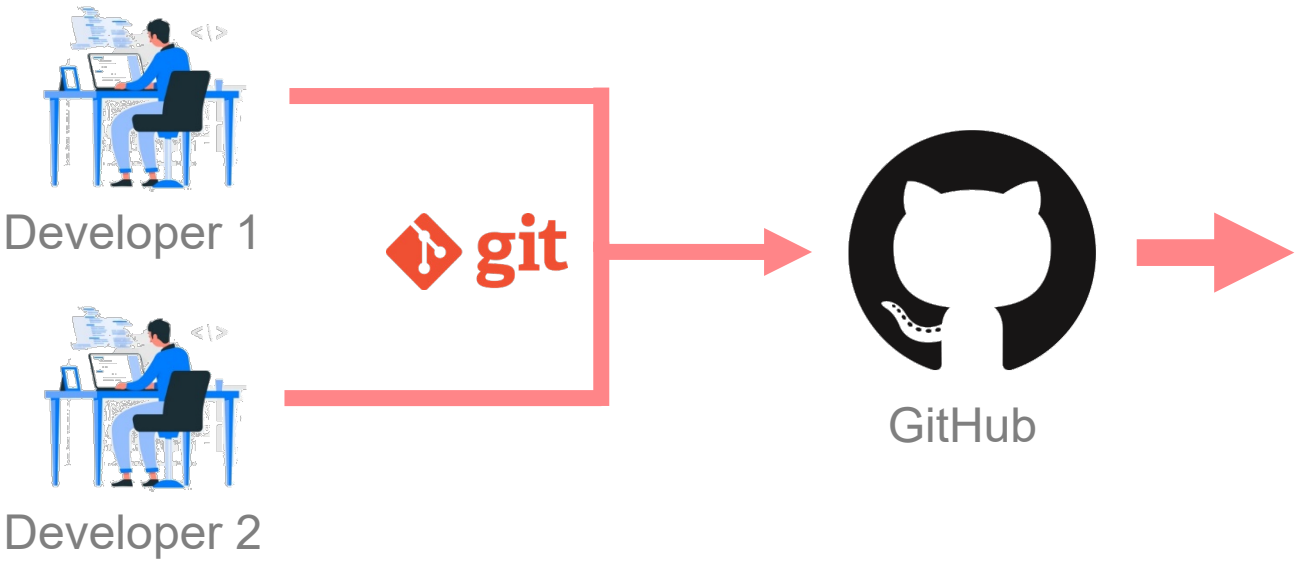
greatlearning
Power Ahead



What is Git

What is Git?

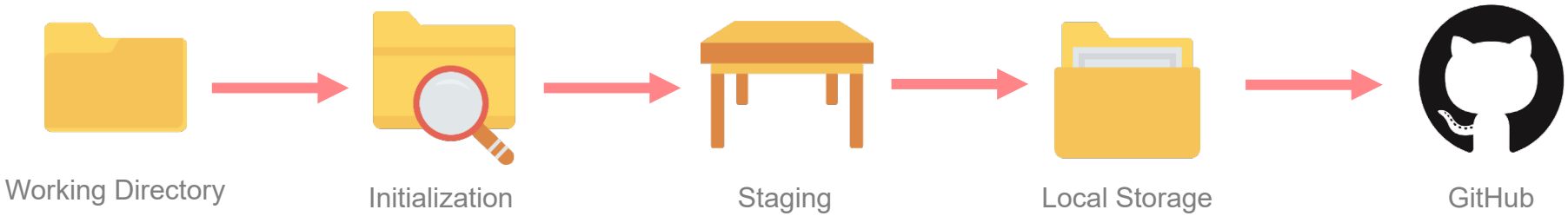
Git is an open source version control system that allows the user to keep track of all the changes that have been made to the source code of the software.



©Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited

What is Git?

The lifecycle of the code within Git.



©Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited

greatlearning
Power Ahead



Getting started with GitHub



Installing & Configuring Git

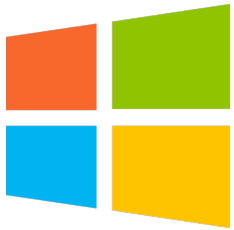
greatlearning
Power Ahead

Git Installation & Setup

Linux



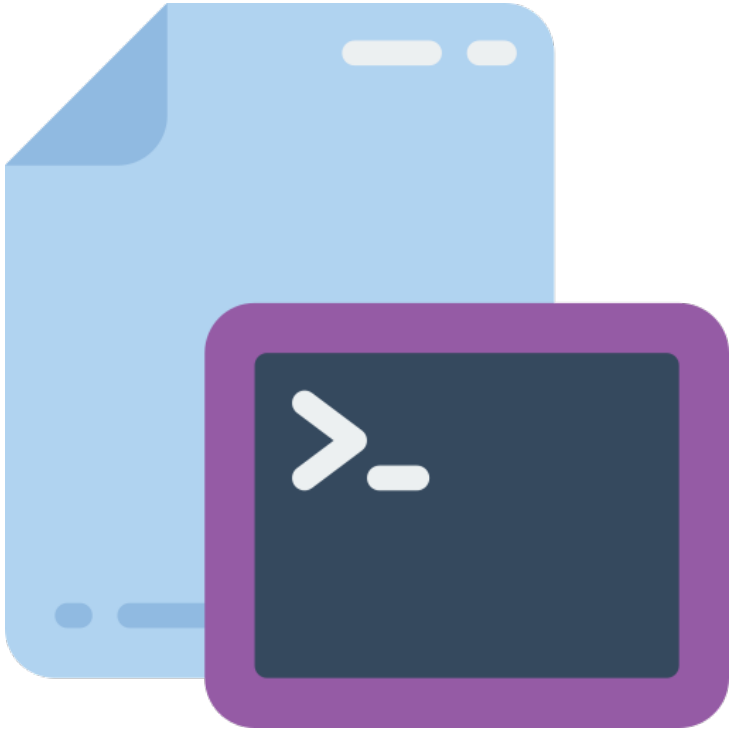
Windows



MAC



greatlearning
Power Ahead



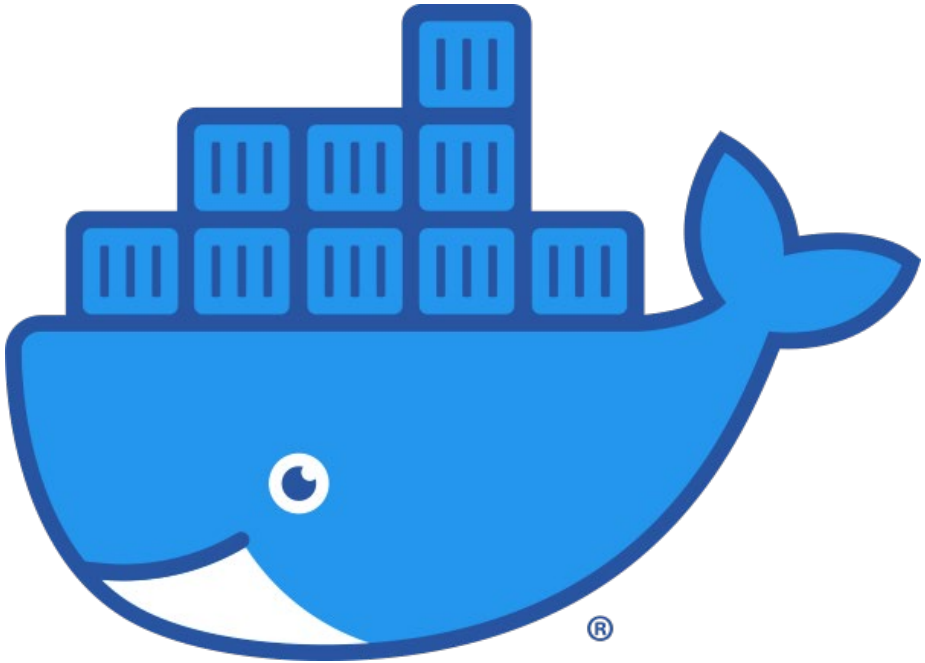
Git Common Commands

greatlearning
Power Ahead



Branches

greatlearning
Power Ahead



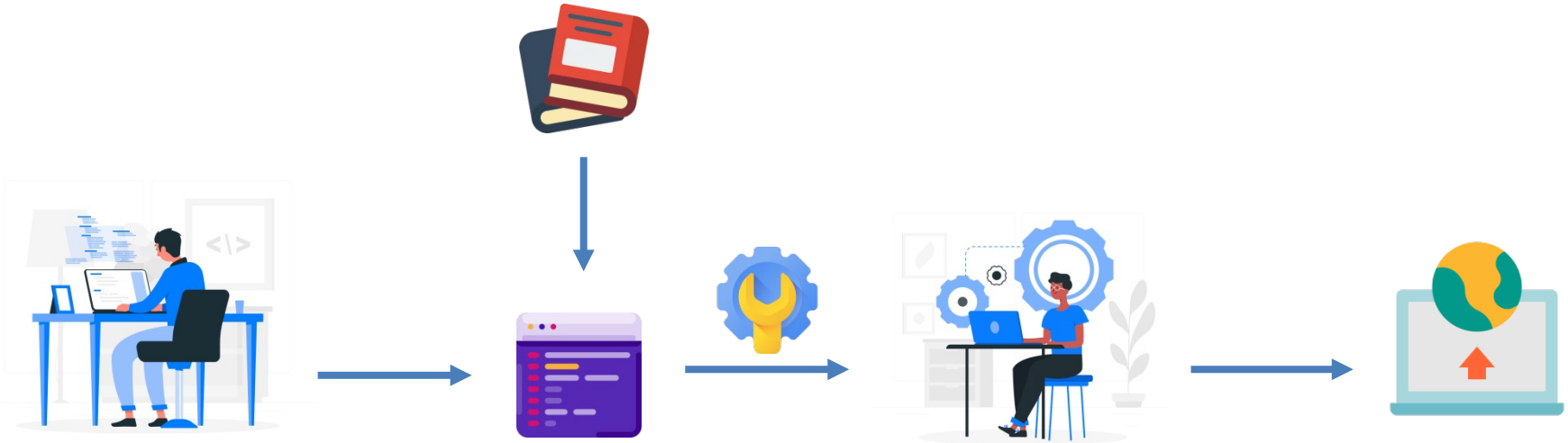
Microsystems & Containerization – Docker



Issues we faced before Containerization

greatlearning
Power Ahead

Issues we faced before Containerization



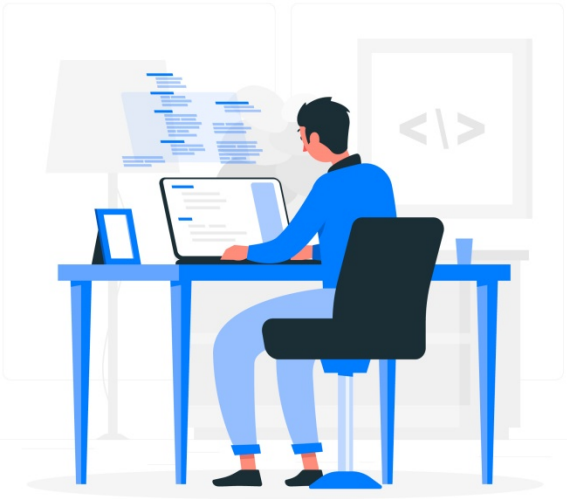
Developer

Tester

Standard Delivery Pipeline

greatlearning
Power Ahead

Issues we faced before Containerization



Backend Developer

Operating System

Pytest 5.4.3

Pycharm IDE

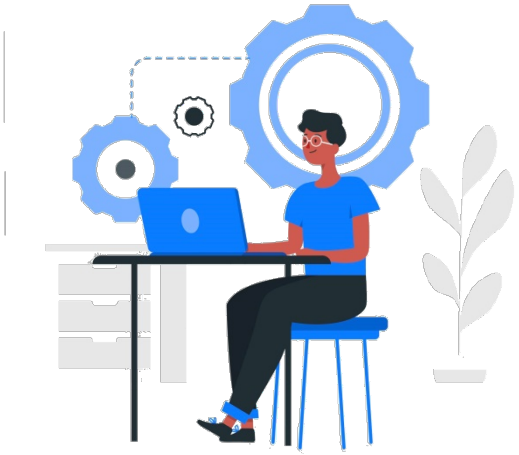
Backend Code

Dependencies

Developer's Environment

greatlearning
Power Ahead

Issues we faced before Containerization



Tester

Operating System

Pytest 5.3.0

Spyder IDE

Backend Code

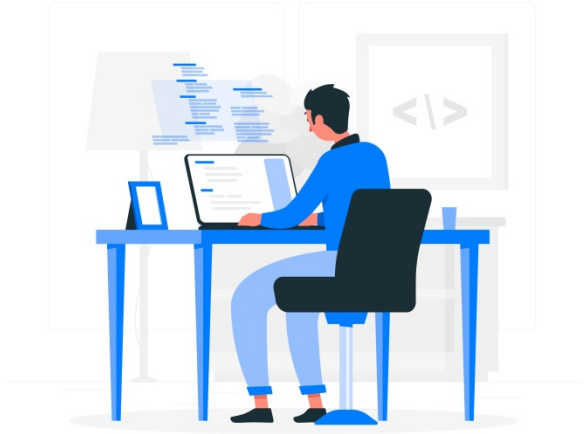
Dependencies

Tester's Environment

greatlearning
Power Ahead

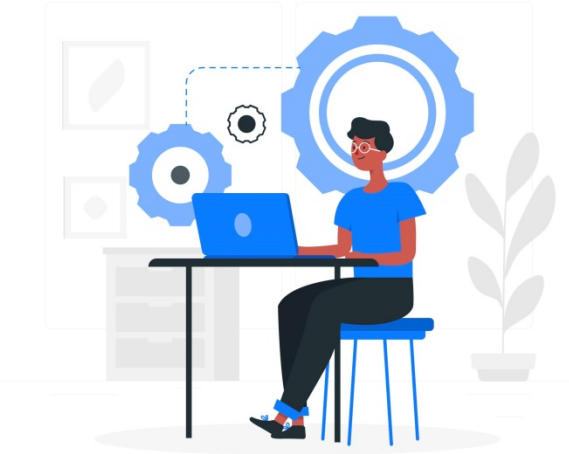
Issues we faced before Containerization

Developer



- This Code runs fine on my computer

Tester



- This Code does not run on my system

greatlearning
Power Ahead

Issues we faced before Containerization

Developer



Operating System



Pytest 5.4.3



Pycharm IDE



Backend Code



Dependencies

Tester



Operating System



Pytest 5.3.0



Spyder IDE



Backend Code



Dependencies

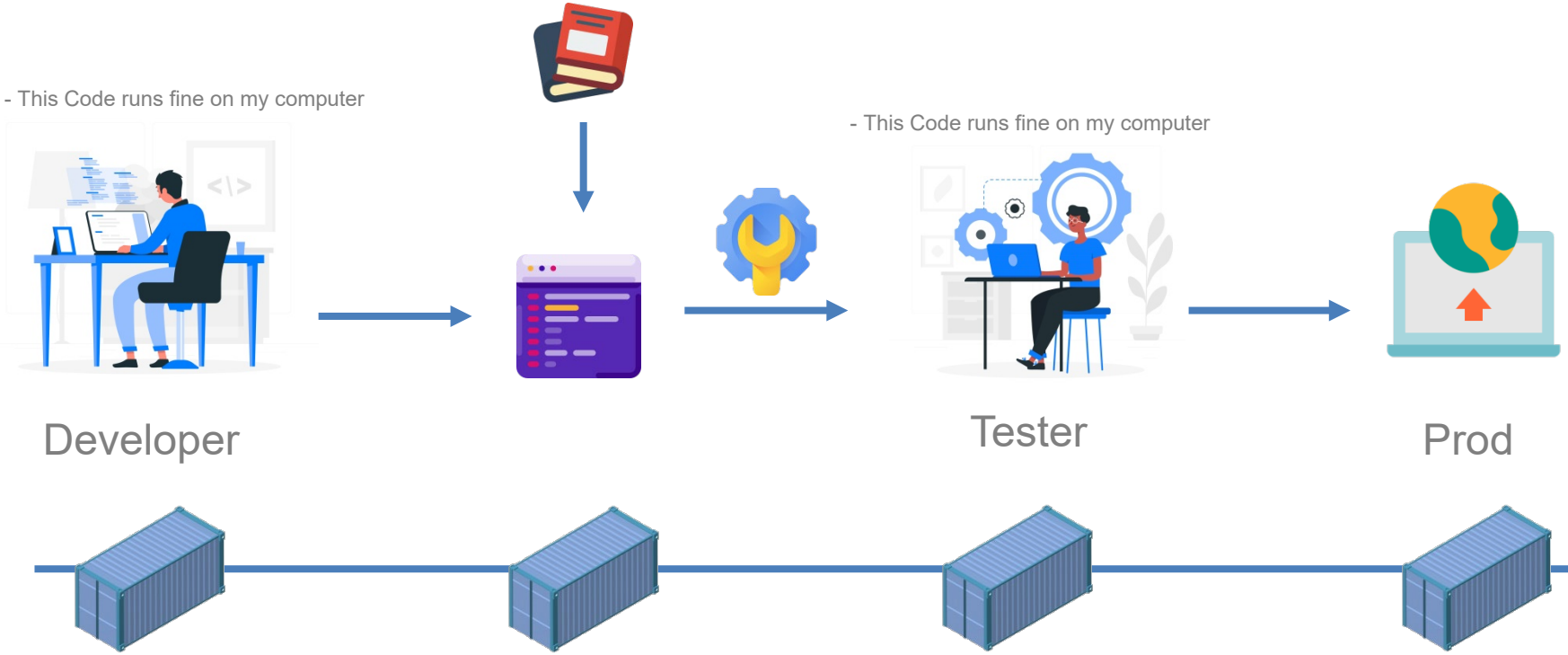
Issues we faced before Containerization



- Now Let's see what happens when we introduce Containers.

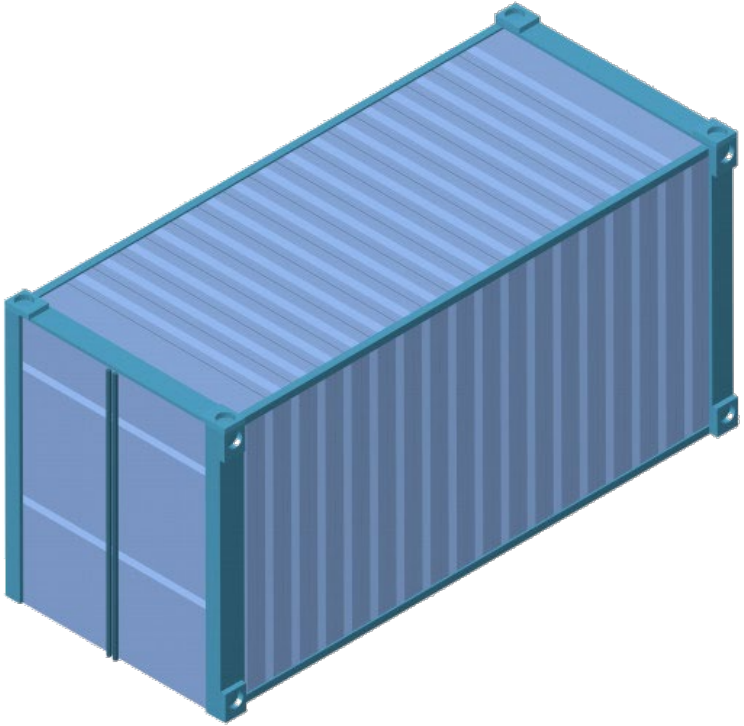
greatlearning
Power Ahead

Issues we faced before Containerization



Similar Delivery Pipeline but with containers

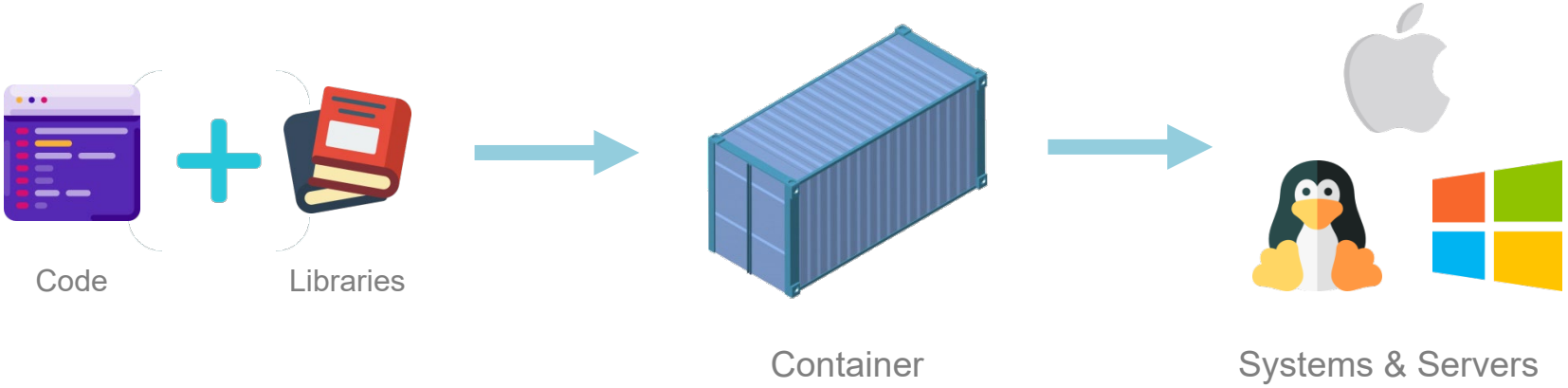
greatlearning
Power Ahead

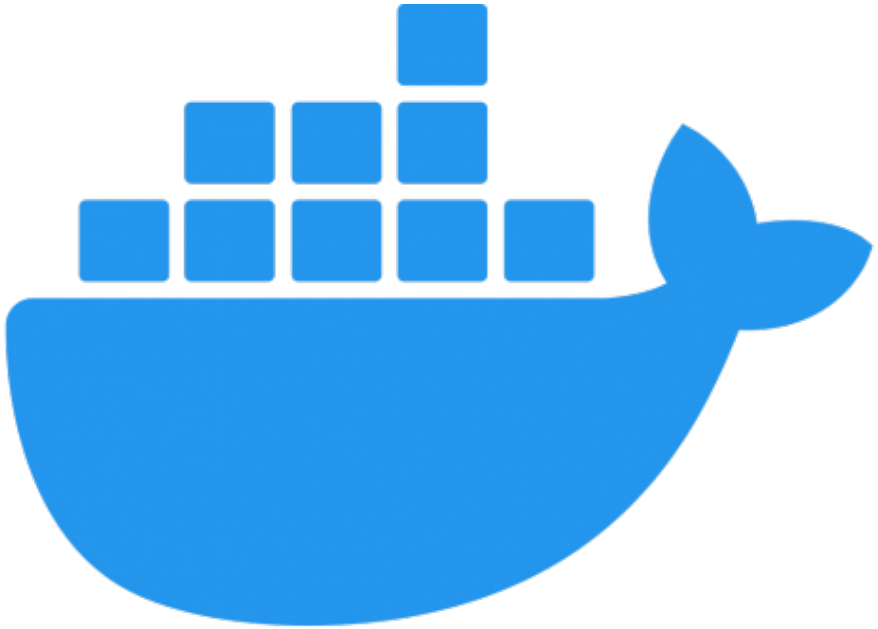


What is a Container?

What is a Container?

Containers are software that wrap up all the parts of a code and all its dependencies into a single deployable unit that can be used on different systems and servers.

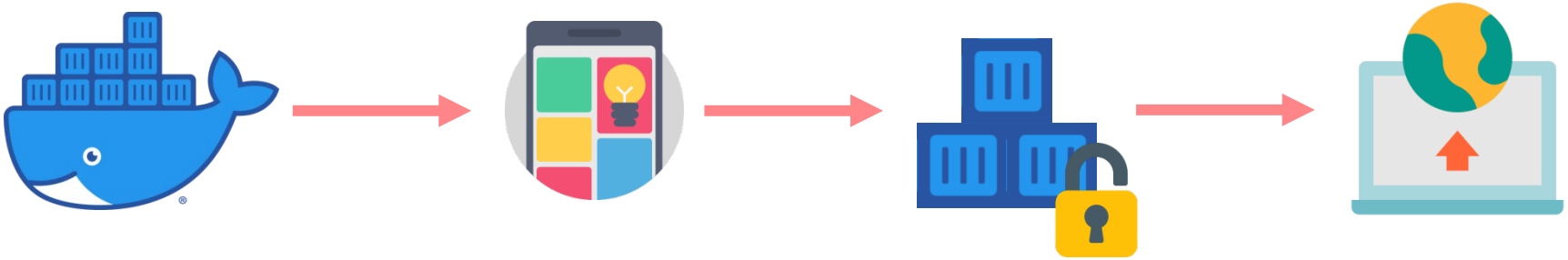




What is Docker?

What is Docker?

Docker is a tool that helps in developing, building, deploying and executing software in isolation. It does so by creating containers that completely wrap a software.



The Isolation provided by container gives a layer of security to the containers.

greatlearning
Power Ahead

Why Docker?



Simple



Fast



Easy Collaboration



Built for Developers, by Developers



Docker Community



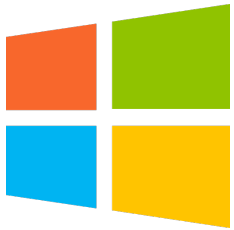
Docker Installation & Setup

Docker Installation & Setup

Linux



Windows



MAC





Docker Environment

greatlearning
Power Ahead

Docker Environment



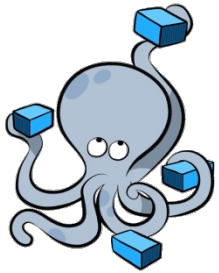
Docker Engine



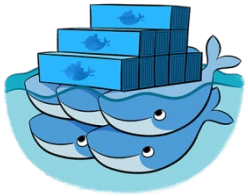
Docker Objects



Docker Registry



Docker Compose



Docker Swarm

greatlearning
Power Ahead



Docker Engine

greatlearning
Power Ahead

Docker Engine

Docker engine is as the name suggests, its technology that allows for the creation and management of all the Docker Processes. It has three major parts to it.



Docker CLI

Docker API

Docker Daemon

greatlearning
Power Ahead



Docker Objects

greatlearning
Power Ahead

Docker Objects



Docker Images



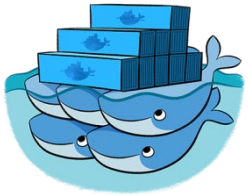
Docker Containers



Docker Volumes



Docker Networks



Docker Swarm Nodes & Services

greatlearning
Power Ahead

Docker Objects - Images

Docker images are sets of instructions that are used to create containers and execute code inside it.

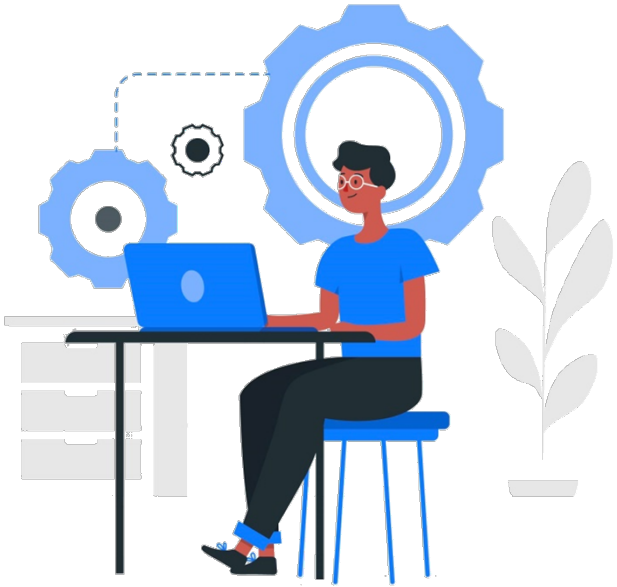


Docker Objects - Images



Let’s pull an image and work with it.

greatlearning
Power Ahead



Docker Common Commands



Continuous Integration - Jenkins

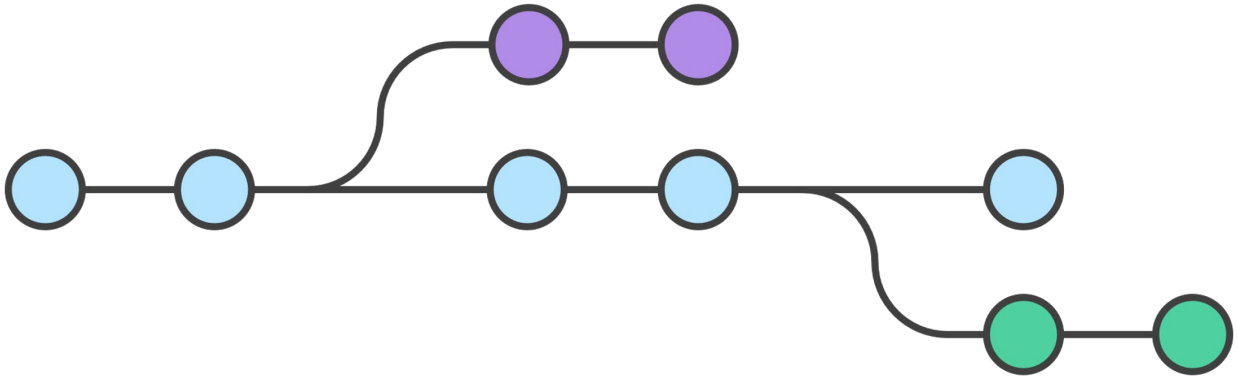
©Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited

greatlearning
Power Ahead

Agenda

1. Why Continuous integration?
2. What is Jenkins?
3. What are Jenkins Features?
4. Jenkins architecture
5. Jenkins installation & setup
6. Setting up a CI/CD pipeline

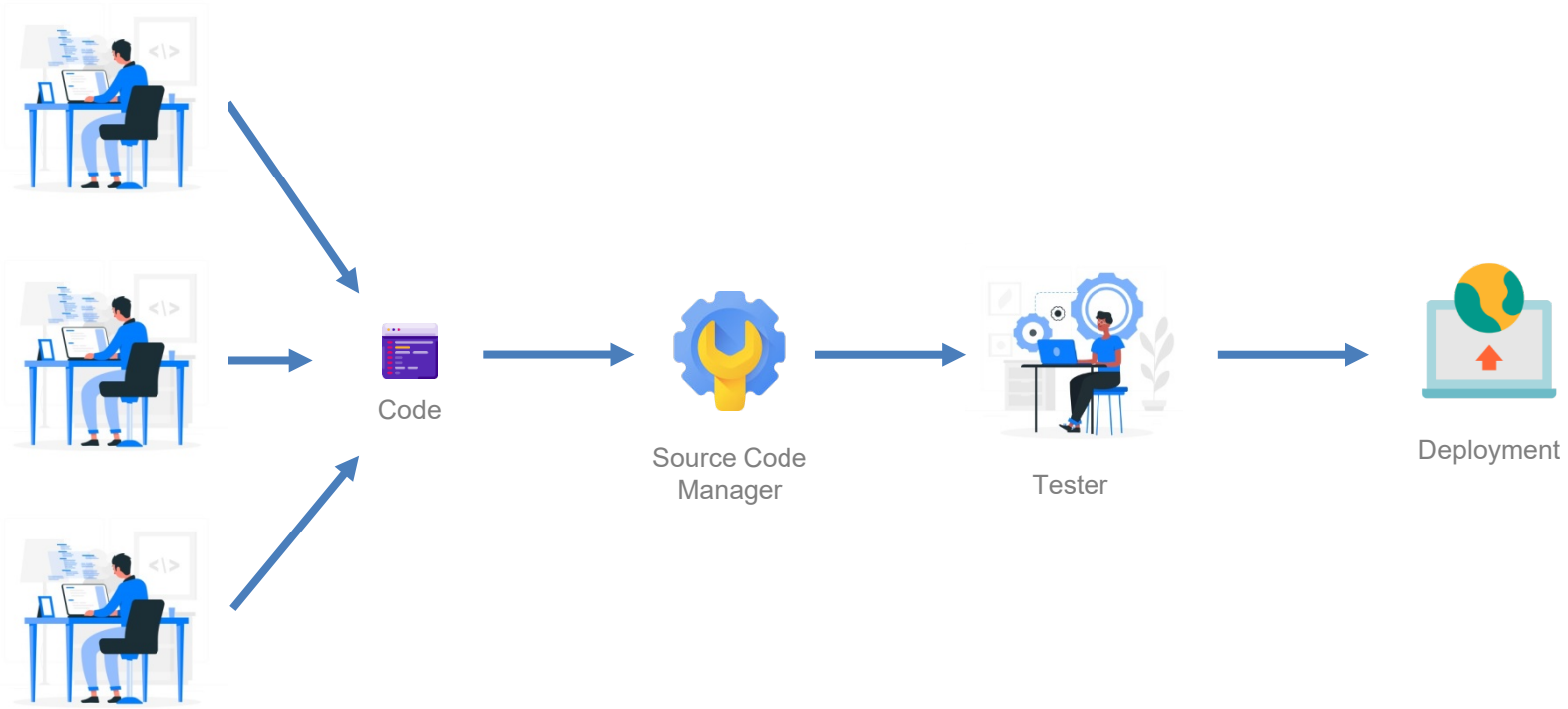
©Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited



Why continuous integration?

©Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited

Why continuous integration?



Standard Delivery Pipeline

©Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited

greatlearning
Power Ahead
Learning for Life



What is Jenkins?

©Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited

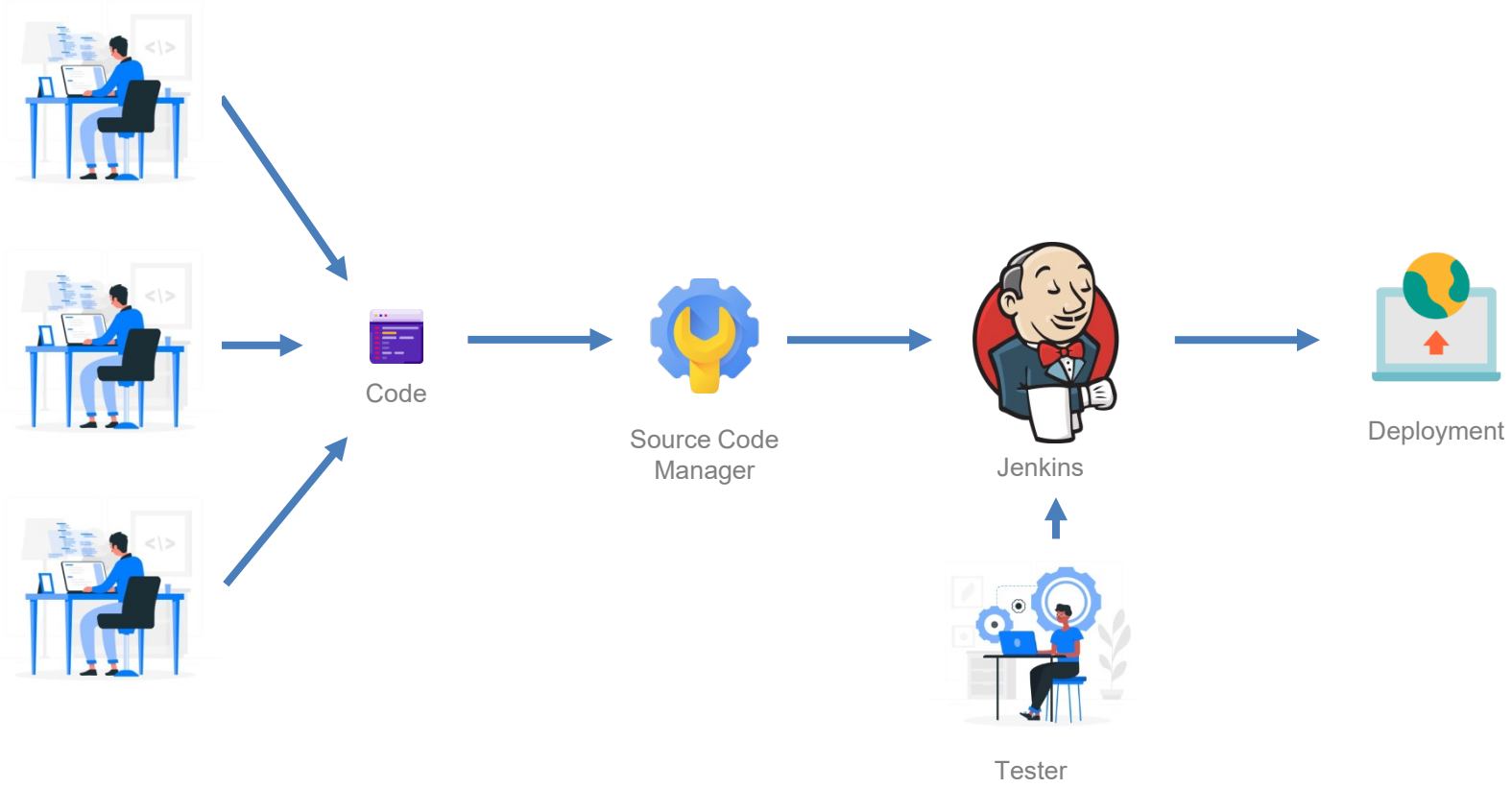
What is Jenkins?

Jenkins is an Open Source continuous integration tool written in java that allows us to automate the software development process, making sure that there is minimum involvement from us. It also integrates all the different parts of the development.



©Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited

What is Jenkins?



©Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited

Benefits of Jenkins

- Jenkins is a community tool – Its open source and has a very very large community that contributes to it regularly.

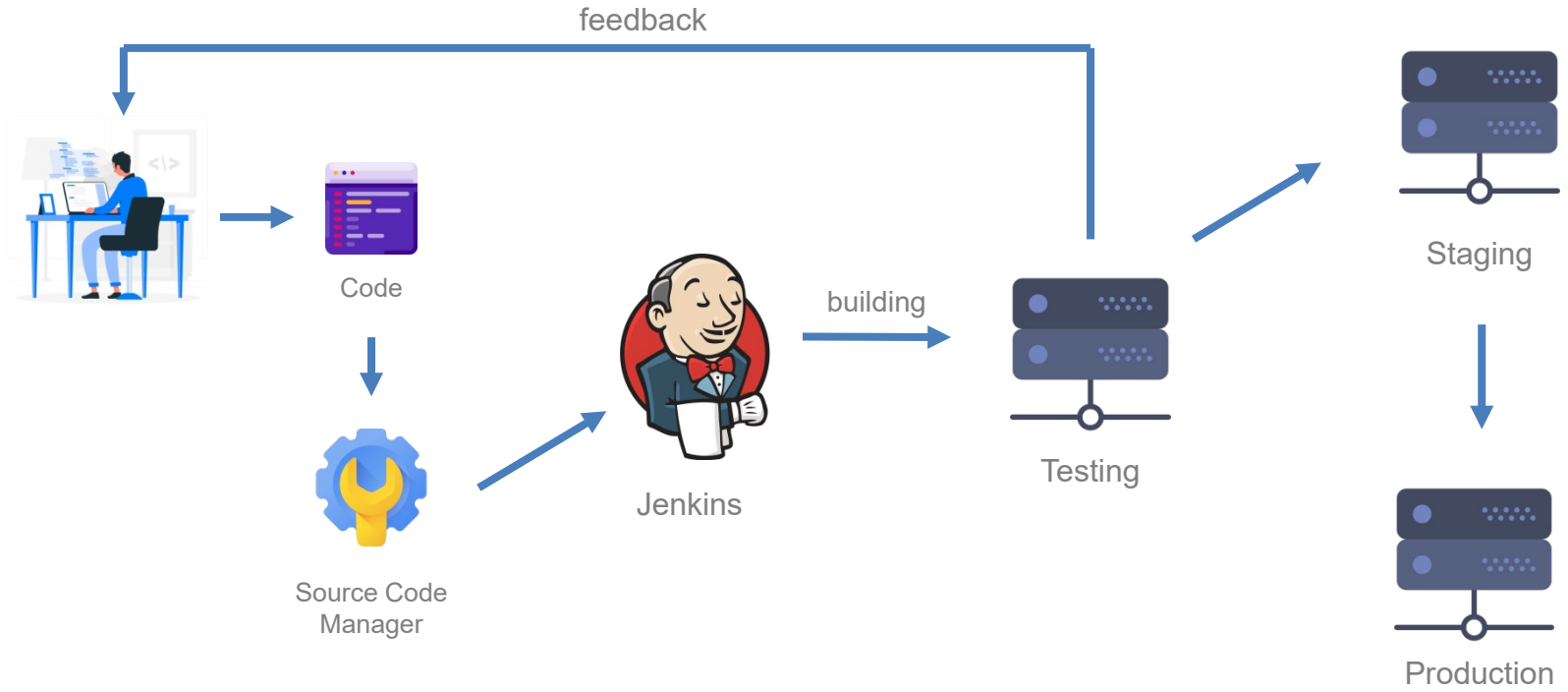
- It's free, so you don't have to accrue anymore subscription costs.

- It is highly modifiable and adaptable and has a support for a very large number of plugins.

- It can run on any major platform without any compatibility issues.

©Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited

Jenkins Pipeline



©Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited

greatlearning
Power Ahead
Learning for Life



Jenkins Installation & Setup

©Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited



Continuous Monitoring – ELK & Prometheus & grafana

©Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited

greatlearning
Power Ahead
Learning for Life



Why Continuous Monitoring?

©Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited

Why Continuous Monitoring?

Continuous Monitoring is an important part of Software Development. It is something we take up as measure to maintain the health of a software and to improve the quality of the software, and this is based on the feed back we get from the insights gained from monitoring.

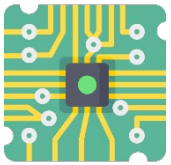


©Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited

Why Continuous Monitoring?

Continuous Monitoring is an important part of Software Development. It is something we take up as measure to maintain the health of a software and to improve the quality of the software, and this is based on the feed back we get from the insights gained from monitoring.













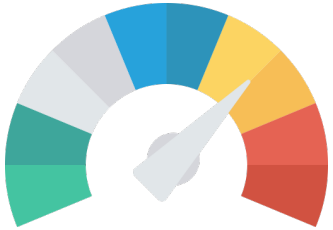
©Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited

Why Continuous Monitoring?

Continuous Monitoring is an important part of Software Development. It is something we take up as measure to maintain the health of a software and to improve the quality of the software, and this is based on the feed back we get from the insights gained from monitoring.







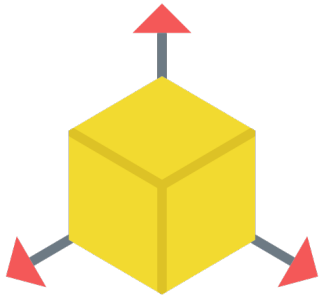
©Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited

Why Continuous Monitoring?

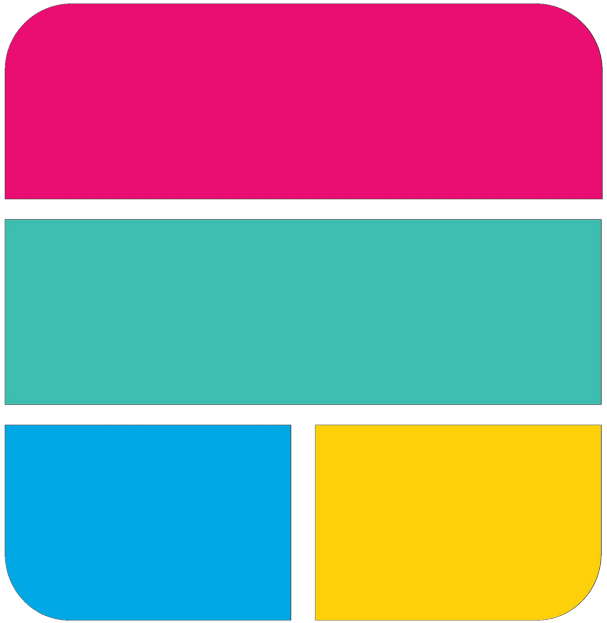
Continuous Monitoring is an important part of Software Development. It is something we take up as measure to maintain the health of a software and to improve the quality of the software, and this is based on the feed back we get from the insights gained from monitoring.







©Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited



What is ELK stack?

©Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited

What is ELK stack?

ELK stack is a set of open-source tools that allow us to monitor, collect, process analyze & visualize data, this data can be of different types and formats and from almost any source. It was developed by Elastic co. iteratively. They started with Elastic search and kept on adding more tools to the stack. The primary purpose of ELK stack is log management.



©Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited

ELK features



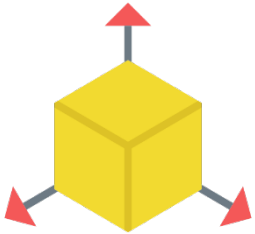
System & Application Performance



Logging



Stack Security & Alerting



Scalability and Resiliency

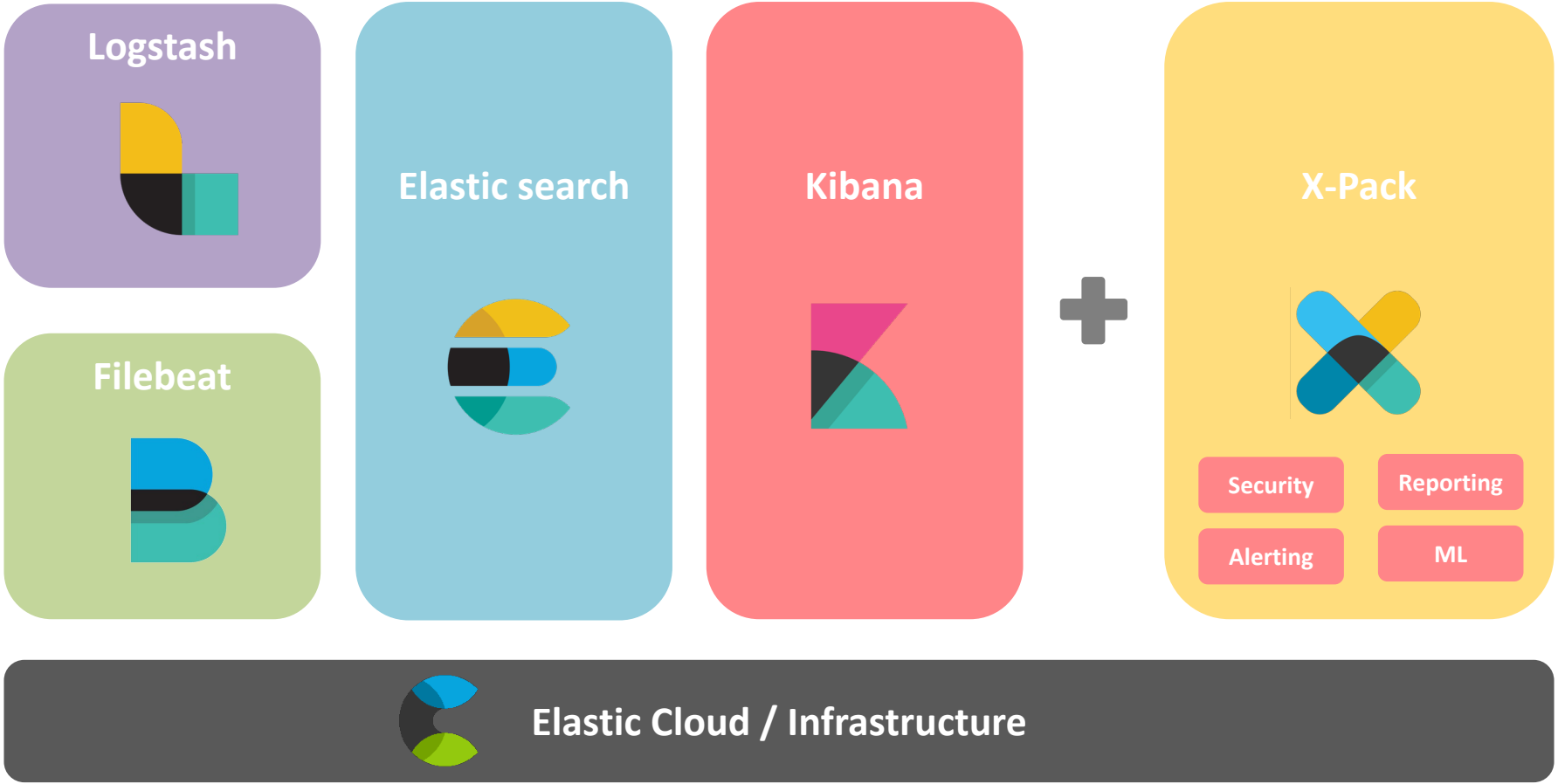


Dashboards & Visualisations

©Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited

greatlearning
Power Ahead

ELK Stack



©Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited



©Great Learning. All Rights Reserved. Unauthorized use or distribution prohibited